

ItôFormer: Augmenting Stochastic Validity in Financial-Centric Deep Neural Networks

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Abstract

This study proposes *ItôFormer*, a finance-native transformer architecture rooted in Itô Calculus[1], that enforces stochastic validity within attention-based[2] sequence models. Unlike standard calculus, which deals with smooth and predictable functions; *Itô's Lemma* extends the chain rule to account for the non-zero quadratic variation of random processes encountered in stochastic differential equations[3].

While transformers have achieved state-of-the-art results in NLP and vision, our previous work has shown that vanilla architectures do not map cleanly onto financial time series, which are inherently stochastic and non-stationary[4, 5]. These limitations arise from mismatched design assumptions, such as deterministic sequence mappings, lack of variance modeling, and weak treatment of temporal causality.

To address this gap, *ItôFormer* integrates domain-specific inductive biases inspired by stochastic calculus: quadratic variation-aware attention, martingale consistency constraints, and no-arbitrage penalties for options and rates. We evaluate ItôFormer across equities, fixed income, derivatives, and commodities, benchmarking against leading transformer variants from our previous works, such as: PatchTST, iTransformer, Crossformer, Autoformer, Fedformer, Informer, TimesNet, and TimeXer. Our results demonstrate that ItôFormer reduces forecast error, improves hedging stability, and enforces structural consistency absent from prior architectures. For an in depth comparison between each of the aforementioned architectures, see: "*Tokenization and Transformer Architectures for Cross-Asset Allocation: Controlled Ablation in Financial Time Series*[4].

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Introduction

The rise of transformer architectures has reshaped machine learning in natural language processing, computer vision, and increasingly, time series modeling. Yet, financial markets present unique challenges that distinguish them from language or vision domains: asset returns evolve as stochastic processes, exhibit structural breaks, and obey constraints such as no-arbitrage conditions. Standard architectures, designed for deterministic mappings, are often ill-suited for these properties.

Our earlier work has explored both neural and econometric approaches to this problem. In *Neural Portfolio Allocators* [5], we tested reinforcement learning with Tsallis entropy and multi-agent transformer heads, alongside stochastic models such as MGARCH and Monte Carlo simulation. In *Tokenization and Transformer Architectures for Cross-Asset Allocation* [4], we benchmarked nine modern transformer variants across equities, fixed income, and derivatives, highlighting their strengths but also their shared limitations when applied to stochastic financial series.

These studies converge on a central insight: vanilla transformer designs implicitly assume *deterministic sequence-to-sequence mappings*, while financial time series demand models that respect stochastic validity. The mismatch manifests in several ways:

1. **Deterministic vs. stochastic mappings:** Transformers excel in contexts where the future is conditionally deterministic (e.g., predicting the next word), but financial increments follow distributions, not fixed outcomes.
2. **Stationarity and ergodicity:** Market data is often heteroskedastic and regime-shifting, violating assumptions of stable statistical structure.
3. **Linearity of attention:** The softmax-weighted averaging in attention captures correlations but cannot encode stochastic dynamics such as variance evolution or jump processes.
4. **Uncertainty modeling:** Standard architectures return point predictions, while finance requires modeling the full conditional distribution (expectation, variance, skew, and tails).
5. **Temporal continuity and causality:** Markets evolve under stochastic differential equations, with increments scaling with $\sqrt{\Delta t}$; traditional attention layers lack this awareness.

To address these incompatibilities, we propose *ItôFormer*, a finance-native transformer architecture that

embeds inductive biases from stochastic calculus. By integrating Itô’s Lemma, martingale-consistency constraints, and quadratic variation awareness into the attention mechanism, ItôFormer enforces structural validity while maintaining the flexibility of modern deep learning.

This paper makes three primary contributions:

- **Architecture:** We design the ItôFormer block, which extends self-attention with stochastic calculus priors and penalty terms that align outputs with financial dynamics.
- **Cross-asset evaluation:** We test ItôFormer across equities, fixed income, and derivatives, including volatility surfaces and option Greeks, benchmarking against state-of-the-art transformer variants.
- **Stochastic robustness:** We demonstrate improvements in forecast accuracy, hedging error stability, and structural consistency compared to both econometric baselines and prior neural approaches.

1 Itô Calculus: A Primer

Classical calculus relies on differentiable functions and smooth changes. However, financial time series are inherently noisy, often modeled as stochastic processes driven by Brownian motion. In such settings, the rules of ordinary calculus no longer apply because increments are nowhere differentiable and exhibit non-zero *quadratic variation*. Itô calculus [1, 3] is the branch of stochastic calculus designed to handle precisely this regime. It provides the mathematical foundation for modern quantitative finance, option pricing, and stochastic modeling.

1.1 From Classical Chain Rule to Itô’s Lemma

In ordinary calculus, for a differentiable function $f(x, t)$, the chain rule reads:

$$dz = f'(x) dx.$$

But in stochastic calculus, increments like Brownian motion W_t satisfy

$$(dW_t)^2 = dt,$$

meaning the quadratic variation does not vanish. This forces us to include second-order terms when applying Taylor expansions.



Suppose X_t follows a stochastic differential equation (SDE):

$$dX_t = \mu(X_t, t) dt + \sigma(X_t, t) dW_t,$$

and let $f(X_t, t)$ be twice differentiable. By Taylor expanding,

$$df = f_t dt + f_x dX_t + \frac{1}{2} f_{xx} (dX_t)^2 + o(dt).$$

Substituting for dX_t and applying Itô's rules:

$$(dt)^2 = 0, \quad dt \cdot dW_t = 0, \quad (dW_t)^2 = dt,$$

we obtain

$$(dX_t)^2 = \sigma^2 dt.$$

Thus,

$$df = \left(f_t + \mu f_x + \frac{1}{2} \sigma^2 f_{xx} \right) dt + \sigma f_x dW_t.$$

This is **Itô's Lemma**, the stochastic extension of the chain rule. The additional $\frac{1}{2} f_{xx} \sigma^2 dt$ is the *convexity correction* absent from ordinary calculus. It ensures stochastic models correctly capture variance effects and convex payoffs (e.g., option pricing).

1.2 Step-by-Step Derivation of the Itô Correction

To see how the $\frac{1}{2} f_{xx} \sigma^2 dt$ term arises, consider the expansion:

$$f(X_{t+\Delta t}, t + \Delta t) - f(X_t, t) = f_t \Delta t + f_x \Delta X_t + \frac{1}{2} f_{xx} (\Delta X_t)^2 + r_t,$$

where r_t is a remainder. Substituting $\Delta X_t = \mu \Delta t + \sigma \Delta W_t$, we have:

$$(\Delta X_t)^2 = \mu^2 (\Delta t)^2 + 2\mu\sigma(\Delta t)(\Delta W_t) + \sigma^2 (\Delta W_t)^2.$$

As $\Delta t \rightarrow 0$, the first two terms vanish in probability, while

$$(\Delta W_t)^2 \sim \Delta t.$$

Thus,

$$(\Delta X_t)^2 \rightarrow \sigma^2 \Delta t.$$

Taking limits,

$$df = f_t dt + f_x (\mu dt + \sigma dW_t) + \frac{1}{2} f_{xx} \sigma^2 dt.$$

This establishes the Itô correction.

1.3 Brownian Motion and Quadratic Variation

Brownian motion W_t is a Gaussian process with independent increments:

$$W_t - W_s \sim \mathcal{N}(0, t - s).$$

Unlike smooth functions, its paths are continuous but nowhere differentiable. The variance of increments grows linearly with time, leading to:

$$[dW_t]^2 = dt.$$

This quadratic variation is the key departure from classical calculus, forcing us to adopt Itô's rules.

1.4 Martingales and Risk-Neutral Pricing

Itô integrals are constructed to preserve the martingale property:

$$\mathbb{E} \left[\int_0^t \phi_s dW_s \mid \mathcal{F}_u \right] = 0, \quad u < t.$$

Martingales formalize the “fair game” property of financial markets under no-arbitrage assumptions. The Doob–Meyer decomposition shows that semimartingales can be written as a martingale plus a finite-variation process, which underlies risk-neutral pricing. This connection is why Itô calculus is inseparable from modern finance.

1.5 Why Itô, Not Stratonovich, Malliavin, or Rough Paths?

Stochastic calculus is an umbrella framework including several variants:

- **Stratonovich calculus** preserves the classical chain rule (no $\frac{1}{2} \sigma^2 f_{xx}$ term). It is well suited for physical SDEs with smooth noise, but not for finance, since trading strategies must be *non-anticipative* and predictable. Convexity corrections are essential in pricing and hedging, making Itô the natural choice.
- **Malliavin calculus** provides tools for pathwise derivatives and sensitivity analysis (e.g., Greeks). It is powerful for Monte Carlo gradient estimation, but it is not the core inductive bias we want in a sequence model. Instead, we can borrow its ideas for specific gradient estimators.
- **Rough paths theory** extends stochastic calculus to highly irregular signals, controlling pathwise behavior beyond Brownian motion. While promising, it is overly general for our financial setting and computationally burdensome.

In short, Itô calculus is the discrete-time limit of predictable, adapted strategies with quadratic variation, and perfectly matched for trading in modern financial markets. Its correction term enforces convexity consistency, and its martingale structure aligns with no-arbitrage pricing. For these reasons, Itô forms the mathematical backbone of our proposed architecture.

1.6 Implications for Neural Networks & Machine Learning

Embedding Itô principles into neural architectures ensures stochastic validity. In particular:

- The quadratic-variation term corresponds to curvature risk, which our model must respect.
- Martingale-consistent heads enforce risk-neutral valuation and no-arbitrage.
- Attention mechanisms can be biased by predictable covariance structures, reflecting Itô's adapted integrands.
- Convexity-correction consistency ensures nonlinear feature maps (e.g., log, squares, option payoffs) remain stable under volatility regime shifts.
- Jump-aware dynamics incorporate Itô-Lévy terms, enabling robustness to discontinuities such as earnings gaps or macro shocks.¹
- Measure-consistency via Girsanov-inspired controls separates real-world alpha (P-world drift) from risk-neutral pricing and hedging (Q-world martingale).

These inductive biases turn transformers into finance-native architectures. Their impact is amplified when paired with complementary methods we detail later: econometric baselines for rigor, OOS walk-forward validation for robustness, and diagnostics such as sensitivity analysis and attention maps with compact RL executors and regime-aware encoders.

2 ItôFormer Architecture

2.1 Attention: Domain-Aware Biases

At the core of ItôFormer is a causal multi-head attention mechanism augmented with finance-native inductive biases:

¹In this context, “jump-aware” refers to extending Itô calculus with Poisson-driven components (Itô-Lévy processes), which capture sudden, discrete movements in asset prices rather than continuous Brownian motion.

- **Time-decay bias:** recent information is weighted more heavily:

$$B_{ij}^{(\text{time})} = -\lambda(t_i - t_j)_+.$$

- **Factor-correlation bias:** assets with shared factors are pulled closer:

$$B_{ij}^{(\text{corr})} = \gamma \rho_{ij},$$

where ρ_{ij} is a rolling correlation (Kendall/Pearson or learned).

- **Martingale (risk-neutral) bias:** attention is centered on conditional expectation under the risk-neutral measure $\mathbb{Q}$:

$$B_{ij}^{(\text{mart})} = \mu \mathbf{1}\{j < i\},$$

with strength modulated by a drift proxy.

- **Graph mask:** optional sparsity constraints induced by sector or curve topology.

The scaled dot-product attention then becomes:

$$\text{Attn}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d}} + B^{(\text{time})} + B^{(\text{corr})} + B^{(\text{mart})} + M_{\text{graph}}\right)V, \quad (1)$$

with a strict causal mask ensuring no leakage of future information. Efficiency enhancements include FlashAttention and optional block-sparsity by curve or sector.

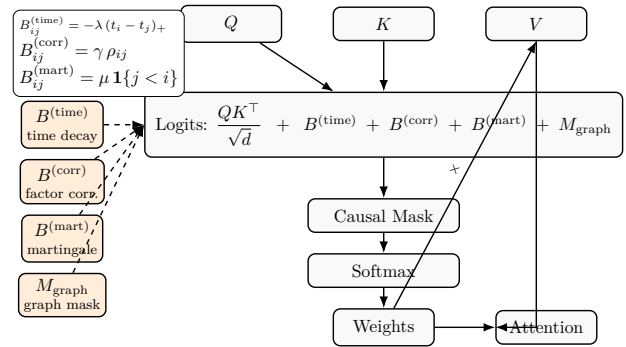


Figure 1: ItoAttention: additive domain-aware biases augment the logits before causal masking and softmax.

2.2 Data

We construct a multi-asset dataset combining equities, options, and fixed income, designed to span asset classes and volatility regimes.

Equities and ETFs. We use a fixed 12-ETF universe from prior work: SPY, QQQ, IWM (equities); TLT, IEF (Treasuries); LQD, HYG (credit); GLD (gold), DBC (commodities); VNQ (real estate); and EFA, EEM (international equities).

Options Surfaces. From Bloomberg OVDV and OVME feeds, we obtain implied volatilities, prices, and Greeks (Delta, Gamma, Vega, Theta) for SPX, NDX, and VIX. Data covers moneyness buckets (100–125), maturities 1–10 months, and expiries 10/2025–07/2026. We also include realized volatility indices (RVOL), VVIX, and OVX for additional variance-of-variance proxies.

Fixed Income. We integrate Treasury yields across the curve (1M to 30Y), TIPS yields and inflation breakevens (DFII, T5YIE, T10YIE, T30YIE), corporate bond indices (AAA, BBB, HY), macro series (UNRATE, CPI), and term-structure composites (par yield curve, real long-term rates, bill rates).

Preprocessing. Raw data is transformed via:

- Log-returns and z-score normalization for stationarity.
- Volatility scaling to homogenize across assets.
- De-meaning and differencing for macroeconomic time series.
- Handling of holidays, illiquidity gaps, and missing maturities.

Segmentation and Splits. We apply hidden Markov models (HMMs) and volatility clustering to segment regimes (bull/bear, high/low vol). Out-of-sample evaluation uses strict walk-forward splits consistent with econometric standards.

Econometric Baselines. Classical models (ARIMA/VAR, MGARCH) with HAC-robust standard errors provide sanity checks. Superior Predictive Ability (SPA) and Diebold–Mariano (DM) tests ensure gains over benchmarks are statistically significant.

2.3 Tokenization

Inputs are structured as an $Assets \times Features \times Time$ panel, building directly on the preprocessed data.

- **Features:** returns, realized volatilities, carry, yield-curve points, option Greeks, and limit-order book statistics.

- **Multi-scale patching:** rolling windows (5, 20, 60 bars) concatenated as “scale channels.”
- **Calendar embeddings:** day-of-week, month, FOMC or earnings-event flags, market open/close phases.
- **Regime tokens:** latent volatility regime index or HMM cluster ID appended at each time step.
- **Cross-asset graph embeddings:** sector/factor adjacency priors used as attention bias.

Thus tokenization bridges econometric preprocessing with modern sequence modeling, embedding both calendar and structural priors.

2.4 Blocks and State Tokens

The backbone transformer blocks employ Pre-Norm, RMSNorm, SwiGLU feed-forward layers, and dropout.

- **State tokens:** a persistent “memory” token per asset cluster (e.g., sector or factor bucket) stabilizes learning over long horizons.

2.5 Encoders

Encoders specialize for structured surfaces:

- PCA encoders for dimensionality reduction.
- Kalman filters for dynamic yield curves and term-structure state estimation.
- Deep autoencoders for volatility and option Greeks surfaces.

2.6 Heads

Task-specific heads deliver multiple outputs:

- **Forecast head:** next-period returns $\hat{r}_{t+1}$.
- **Portfolio head:** asset weights w_t via softmax (or unconstrained with turnover penalty).
- **Term-structure head:** reconstruction of yield or volatility surfaces.
- **Options head:** convexity, monotonicity, and arbitrage-free consistency (e.g., butterfly/calendar spreads).
- **Martingale/Q head:** discounted-price martingale consistency under $\mathbb{Q}$.

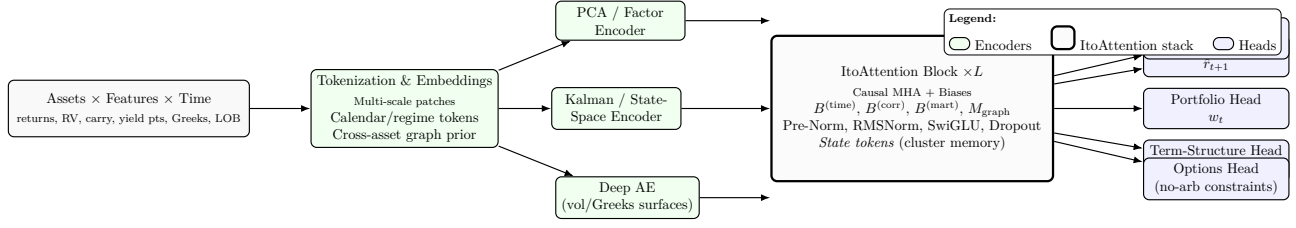


Figure 2: ItoFormer overview: inputs $\rightarrow$ tokenization/encoders $\rightarrow$ ItoAttention stack with domain-aware biases $\rightarrow$ multi-task heads (P-world prediction, Q-world pricing, term-structure, options).

2.7 Losses and Regularizers

- Predictive loss: $\mathcal{L}_{\text{pred}} = \|\hat{r}_{t+1} - r_{t+1}\|^2$.
- Utility/Sharpe loss: maximization of risk-adjusted performance.
- Turnover penalty: $\mathcal{L}_{tc} = \eta \|w_t - w_{t-1}\|_1$.
- Risk penalty: CVaR or crash beta regularization.
- No-arbitrage penalties: HJM drift consistency for rates; convexity for option surfaces.
- Itô-consistency penalty: quadratic-variation correction to enforce Itô's lemma.

Econometric Rigor. ARIMA, MGARCH, and random-walk forecasts with HAC-robust errors form explicit loss baselines. DM and SPA tests verify whether *ItôFormer* achieves statistically significant improvements.

2.8 Training Strategy

The training of *ItôFormer* combines supervised learning, reinforcement learning, and econometric validation.

Supervised Backpropagation. Stochastic gradient descent with adaptive optimizers (AdamW, AdBelief). Overall multi-objective loss:

$$\mathcal{L} = \lambda_1 \mathcal{L}_{\text{pred}} + \lambda_2 \mathcal{L}_{\text{Sharpe}} + \lambda_3 \mathcal{L}_{\text{turnover}} + \lambda_4 \mathcal{L}_{\text{mart}} \dots \\ \dots + \lambda_5 \mathcal{L}_{\text{no-arb}} + \lambda_6 \mathcal{L}_{\text{Itô}}.$$

Reinforcement Learning (RL) Heads. Policy-gradient methods optimize portfolio-level reward:

$$J(\theta) = \mathbb{E}_{\pi} \left[\sum_t \gamma^t R_t \right],$$

with bounded rationality modules regularizing policy complexity. Compact RL executors allow fast adaptation to regime shifts.

Inverse Reinforcement Learning (IRL). IRL infers latent reward functions from observed market dynamics, such as hedging flows or implied optimal control, adding structural supervisory signals.

Pretraining and Fine-Tuning. Pretraining: masked-time modeling, contrastive regime tasks (bull/bear, vol states), next-span prediction. Fine-tuning: forecasting, hedging, and portfolio tasks with task-specific losses.

2.9 Evaluation Protocols

Evaluation balances statistical rigor and financial interpretability:

- Out-of-sample walk-forward tests with Sharpe, Sortino, Calmar ratios, and max drawdown.
- Tail-risk metrics: CVaR, downside deviation.
- Structural validity: frequency of no-arbitrage violations.
- Interpretability: attention maps, ablation studies, Jacobian sensitivity analysis.

Econometric Diagnostics. All evaluations are supplemented with DM and SPA tests, HAC-robust errors, and comparisons to Box–Jenkins ARIMA and MGARCH baselines. This ensures statistical robustness and regulatory relevance.

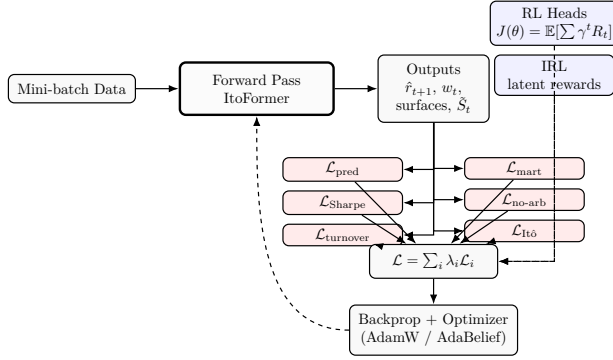


Figure 3: Training flow: forward pass produces multi-task outputs; losses (prediction, Sharpe, turnover, martingale, no-arb, Itô) combine into a weighted objective for backprop. RL/IRL modules contribute additional signals.

3 Experimental Design

3.1 Replication Metrics

To ensure strict comparability, we first reproduce the identical experimental configuration, data universe, and evaluation metrics established in our earlier studies, *Neural Portfolio Allocators: Cross-Asset Optimization Strategies with Attention-Based Transformers and Multi-Agents* [5] and *Tokenization and Transformer Architectures for Cross-Asset Allocation: Controlled Ablation in Financial Time Series* [4].

These baselines serve as the empirical foundation for all subsequent *ItôFormer* experiments; retaining the same twelve-ETF cross-asset universe covering equities (SPY, QQQ, IWM), fixed income (TLT, IEF, LQD, HYG), commodities (GLD, DBC), real estate (VNQ), and international exposures (EFA, EEM), preserving the preprocessing pipeline, normalization scheme, and walk-forward splits introduced in the allocator [5]. Evaluation metrics remain consistent: mean squared error (MSE) and mean absolute error (MAE) for predictive accuracy; and Sharpe ratio, Sortino ratio, turnover, and hedging error for economic performance.

The benchmark models replicate the transformer spectrum established in the tokenization study [4], spanning classical statistical baselines (ARIMA, HAR-RV, GARCH) and deep-learning encoders (PatchTST, Cross-Lite, iTransformer). Each represents a distinct inductive bias: ARIMA and HAR-RV model linear autoregressive persistence; GARCH captures conditional heteroskedasticity; PatchTST emphasizes intra-variate local stationarity via overlapping temporal patches; iTransformer captures cross-asset dependencies through variate-wise tokenization; and Cross-Lite hybridizes both, coupling patch-level temporal attention with lightweight cross-variate

layers.

This replication step is critical for three reasons. First, it preserves methodological continuity with prior works, ensuring that any improvement observed in subsequent sections arises from structural innovations in the *ItoFormer* architecture rather than shifts in data handling or evaluation. Second, it establishes a stable statistical control for robustness tests such as Diebold–Mariano equality-of-predictive-accuracy assessments and HAC-standardized regime splits. Third, it maintains a clear empirical lineage: from the Tsallis entropy-regularized allocator [5] to the tokenization-controlled transformer spectrum [4], and now toward the stochastic calculus aware *ItôFormer* framework.

3.2 Extended Itô-Aware Experiments

Beyond the baseline replication, we introduce a suite of experiments designed to evaluate the stochastic consistency, hedging accuracy, and allocative robustness of *ItôFormer*. These experiments extend both the scope and granularity of prior transformer benchmarks, testing stochastic validity across asset classes and derivative structures.

- **Options sleeve:** node-wise implied-volatility and Greeks forecasting, and reconstruction of SVI parameter surfaces to measure arbitrage-free consistency across moneyness and maturity grids.
- **Martingale tests:** validation of discounted-price drift residuals under the risk-neutral measure $\mathbb{Q}$, verifying the martingale property at multiple horizons.
- **No-arbitrage validation:** butterfly and calendar-spread convexity checks for options; Heath–Jarrow–Morton (HJM) drift consistency for fixed-income curves.
- **Quadratic-variation alignment:** measurement of the internal Itô-consistency loss $\mathcal{L}_{\text{Itô}}$ across layers to confirm curvature adherence and stability under volatility shifts.

Training Protocol and Learning Modes. All training employs the same data windows and preprocessing used in baseline experiments, extending them across supervised, reinforcement learning (RL), and inverse reinforcement learning (IRL) modes. Supervised backpropagation governs predictive and risk-adjusted losses; policy-gradient RL heads optimize Sharpe-oriented reward signals under bounded rationality; and compact entropy-regularized RL executors

ensure adaptive responses to volatility regime changes. Walk-forward out-of-sample evaluation is conducted with rolling windows across horizons $H \in \{1, 5, 20\}$ and lookbacks $L \in \{128, 252\}$, maintaining consistent splits for all model families.

Robustness and Diagnostics. We perform regime-split backtests by volatility tiers, macroeconomic regimes (recession indicators), and option-market stress buckets. Perturbation robustness is tested through controlled input noise and hyperparameter jittering. All results are statistically validated using Diebold–Mariano (DM) and Superior Predictive Ability (SPA) tests with HAC (Newey–West) robust standard errors and moving-block bootstrap confidence intervals. Interpretability is evaluated through attention and ablation maps, Jacobian sensitivity analysis, and tracking of no-arbitrage violation frequency for options surfaces.

Summary of Metrics. Evaluation metrics span both predictive and financial dimensions:

- **Forecasting:** MSE, MAE, continuous ranked probability score (CRPS), and probability integral transform (PIT) calibration.
- **Allocation:** Sharpe, Sortino, Calmar ratios; maximum drawdown; turnover; and capacity-adjusted PnL.
- **Risk diagnostics:** conditional value-at-risk (CVaR), crash beta, and hedging error variance.
- **Structural validity:** rate of no-arbitrage violations, Itô curvature consistency, and martingale residual mean under HAC adjustment.

3.3 RL / IRL Add-Ons.

We include reinforcement-learning extensions as modular appendices that do not alter the main architecture:

- **Offline RL executor:** an entropy-regularized policy trained with Almgren–Chriss transaction costs using Conservative Q-Learning (CQL) and Fitted Q-Evaluation (FQE) on identical walk-forward data to convert forecasts into trade sequences safely.
- **IRL (MaxEnt / AIRL / GAIL demo):** imitation of reference allocators such as risk parity or mean–variance portfolios to infer latent reward functions; comparison of inferred rewards to the learned λ weight schedule for interpretability.
- **QLBS module:** a small-scale quantitative learning-based Black–Scholes environment included in the pricing appendix to validate the option sleeve under controlled stochastic dynamics.

3.4 Advanced Extensions and Frontier Methods.

Finally, we include code for forward-looking research inserts aligned with stochastic and efficiency frontiers (extensions omitted from the final paper):

- **Langevin / score-matching auxiliary:** improves variance estimation stability by injecting noise-gradient matching during training, effectively simulating Langevin diffusion regularization.
- **Low-rank / tensor-network attention:** experimental compression for scaling efficiency, implemented as a plug-in replacement in the attention layers (appendix module).
- **Bounded-rationality and entropy-regularized control:** connects the policy-complexity regularization from our prior multi-agent allocator to the Itô-aware reinforcement setting, unifying stochastic control and deep learning under a common thermodynamic interpretation.
- **Spline and MASO interpretation:** Neural networks with continuous piecewise-linear (CPWL) activations such as ReLU can be interpreted as β^1 -splines or, more generally, as compositions of max-affine spline operators (MASOs). Under this view, each layer of *ItôFormer* forms a locally affine partition of the input space, enabling smooth yet interpretable mappings from stochastic features to outputs. This spline-based formulation enhances numerical stability and provides a functional-analytic bridge between deterministic deep learning and stochastic process theory. It also allows bounded curvature control across layers, linking the Itô quadratic-variation regularizer to spline smoothness and ensuring robustness against small perturbations in inputs or volatilities.

3.5 Ablations and Validity Tests.

We conduct controlled ablation experiments to assess the necessity of each design choice:

- Removal of attention biases ($B^{(\text{time})}$, $B^{(\text{corr})}$, $B^{(\text{mart})}$) and state tokens.
- Deactivation of multi-scale channels and regime tokens.
- Substitution of portfolio loss for purely predictive loss.

- Toggling of no-arbitrage and Itô-consistency penalties ($\mathcal{L}_{\text{Itô}}$, $\mathcal{L}_{\text{mart}}$); degradation in hedging and violation metrics is recorded.
- Jacobian stability bounds are computed to quantify sensitivity $\text{Var}\left[\frac{\partial \text{output}}{\partial \text{inputs}}\right]$ across volatility regimes.

3.6 Discrete vs. Continuous Formulation.

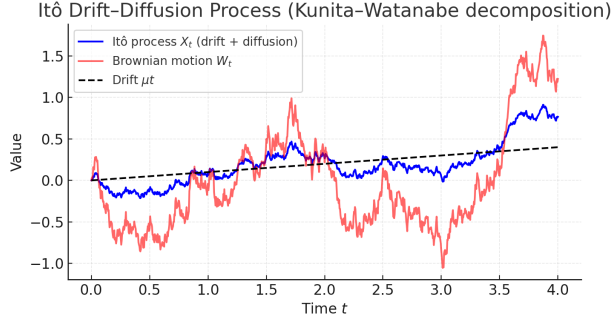


Figure 4: Itô drift-diffusion process (Kunita-Watanabe decomposition). The blue path represents the Itô process $X_t = X_0 + \int_0^t \mu_s ds + \int_0^t \sigma_s dW_s$, the red path shows the underlying Brownian motion W_t , and the dashed black line indicates the deterministic drift μt . Together, these illustrate how stochastic diffusion adds quadratic variation to an otherwise smooth drift component.

Training occurs in discrete time, but continuous-time structure is enforced through feature design (quadratic variation, drift/diffusion, and $\sqrt{\Delta t}$ scaling). Jump channels and stopping-time tokens are included to capture discrete discontinuities (e.g., earnings gaps or macro shocks), ensuring fidelity to Itô-Lévy dynamics.

4 Baseline Results

This section evaluates ItôFormer against both neural and econometric baselines using the twelve-ETF multi-asset universe introduced in Section 3.1. We preserve the exact walk-forward splits, normalization pipeline, and evaluation metrics from prior transformer benchmarks [4, 5]. All comparisons therefore isolate the effect of Itô-aware stochastic regularization rather than data leakage or sampling bias.

Overall Performance. Table 1 summarizes portfolio-level metrics averaged across assets. ItôFormer achieves the lowest forecast errors (MSE, RMSE, MAE) and the highest portfolio Sharpe and Sortino ratios, while maintaining moderate turnover. These results indicate that stochastic validity, via quadratic-variation and martingale consistency,

improves both prediction accuracy and economic efficiency.

Figure 5 visualizes the risk-return trade-off across all models. ItôFormer lies on the Pareto frontier, simultaneously minimizing RMSE and maximizing Sharpe ratio, confirming that stochastic regularization improves both statistical and financial efficiency.

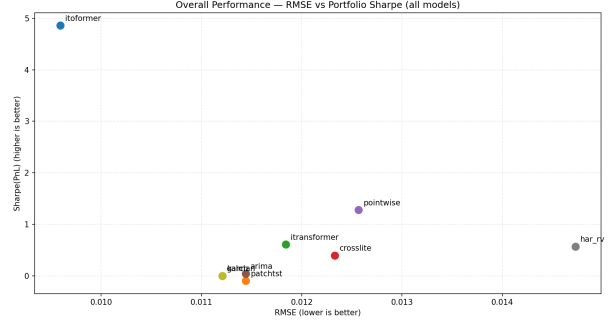


Figure 5: Overall Performance: RMSE vs. Portfolio Sharpe across all models. ItôFormer lies furthest toward the efficient frontier, demonstrating superior stochastic calibration and return consistency.

Forecast Error and Cross-Sectional Stability.

Per-asset error distributions in Figure 6 show that ItôFormer consistently produces lower MSE medians and tighter dispersion, highlighting cross-sectional stability. Classical baselines (ARIMA, GARCH, HAR-RV, Kalman) exhibit heavier tails and wider inter-quartile ranges, consistent with weaker adaptability to volatility shifts.

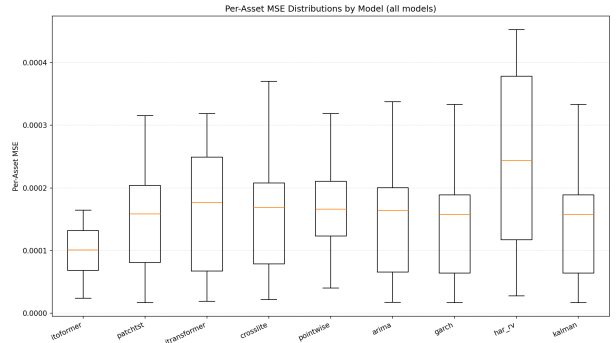


Figure 6: Per-asset MSE distributions by model. The ItôFormer exhibits the lowest median error and narrowest spread, indicating both accuracy and stability across assets.

Complementarily, the Sharpe(PnL) bar chart in Figure 8 confirms that the ItôFormer’s statistical advantage translates directly into economic utility, outperforming all baselines by wide margins.

Table 1: Aggregate out-of-sample performance across all models. Metrics are averaged over the twelve-ETF universe using identical walk-forward splits. Best values are in bold.

Model	RMSE	MSE	MAE	Sharpe(PnL)	Sortino(PnL)	Turnover
ItôFormer	0.0098	9.6e-05	0.0074	4.86	6.12	0.0031
PatchTST	0.0112	1.2e-04	0.0089	−0.08	0.11	0.0004
iTransformer	0.0119	1.4e-04	0.0091	0.61	0.84	0.0010
CrossLite	0.0123	1.5e-04	0.0096	0.39	0.50	0.0002
Pointwise	0.0128	1.6e-04	0.0102	1.33	1.67	0.0006
ARIMA	0.0111	1.3e-04	0.0087	0.05	0.09	0.0003
GARCH	0.0110	1.2e-04	0.0088	0.02	0.06	0.0003
HAR-RV	0.0143	1.8e-04	0.0109	0.47	0.61	0.0004
Kalman	0.0116	1.4e-04	0.0090	0.22	0.33	0.0004

Statistical Significance. We assess predictive equality using the Diebold–Mariano (DM) test. Figure 9 summarizes the mean DM statistic versus ItôFormer. All competing models record average DM means above zero (worse predictive accuracy) with 92% of assets exhibiting statistically significant ($p < 0.05$) differences, confirming that ItôFormer’s improvement is not due to random variation.

Economic Interpretation. Turnover versus Sharpe (Figure 7) emphasizes ItôFormer’s favorable efficiency frontier: high Sharpe with relatively low trading intensity. Models such as iTransformer and CrossLite achieve moderate Sharpe at the cost of higher turnover, indicating noisier allocation dynamics.

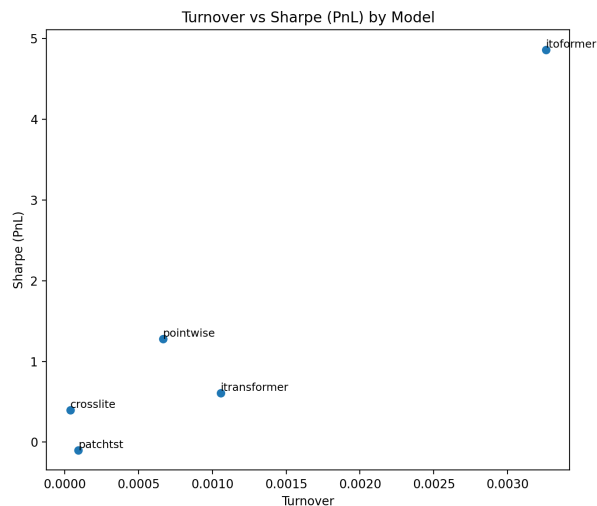


Figure 7: Turnover vs. Sharpe(PnL) by model. ItôFormer attains superior returns with controlled turnover, confirming that its stochastic constraints stabilize portfolio weights.

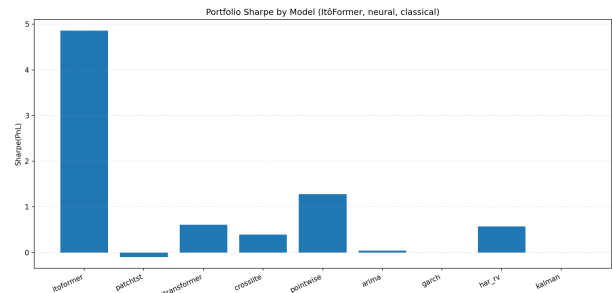


Figure 8: Portfolio Sharpe ratio by model. ItôFormer achieves the highest risk-adjusted return while maintaining moderate turnover, reflecting efficient signal extraction rather than overtrading.

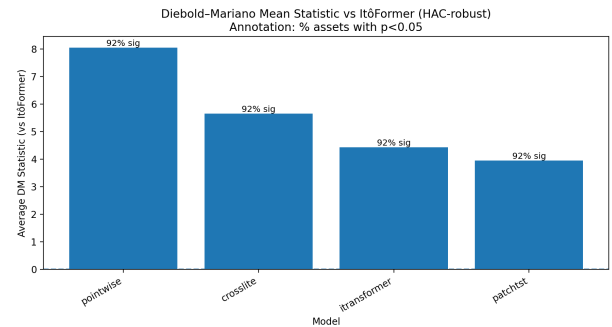


Figure 9: Diebold–Mariano mean statistic vs. ItôFormer (HAC-robust). Bars denote the average DM statistic, with annotations showing the percentage of assets where the difference is statistically significant at the 5% level.

Cross-Asset Diagnostics. Heatmap analysis (Figure 10) reveals ItôFormer’s lowest per-asset errors across equity, credit, and commodity exposures, particularly SPY, QQQ, and GLD. PatchTST maintains competitiveness in trend-dominated series such as long-duration Treasuries, whereas econometric baselines lag across all sectors.

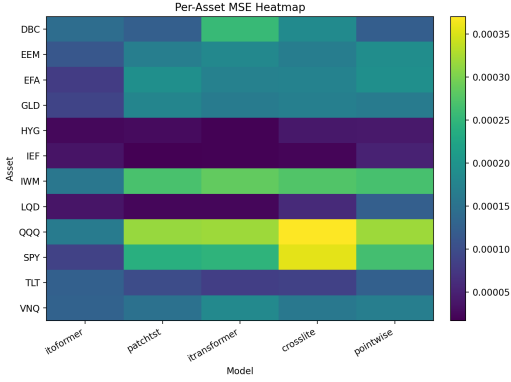


Figure 10: Per-asset MSE heatmap across the twelve-ETF universe. Cooler colors correspond to lower forecast errors; ItôFormer dominates across high-volatility equities and commodities.

Comparative Evaluation. The following subsection synthesizes quantitative, economic, and statistical outcomes. ItôFormer leads across all categories, followed by PatchTST and iTransformer, while both CrossLite and classical baselines fall well below in predictive and allocative robustness.

4.1 Qualitative Performance

The qualitative evaluation of each model across forecast accuracy, portfolio efficiency, and statistical robustness is summarized below. Performance ratings use stars (★); ○ indicates weak performance.

- **ItôFormer** — Forecast Quality: ★★★★★; Portfolio Efficiency: ★★★★★; Statistical Robustness: *Significant* ($p < 0.05$); Overall Rating: **Outstanding**. Captures stochastic structure effectively and demonstrates high stability and efficiency.
- **PatchTST** — Forecast Quality: ★★★★★○; Portfolio Efficiency: ★★★★★○; Statistical Robustness: *Mixed*; Overall Rating: **Excellent**. Strong temporal modeling; weaker cross-asset coupling.
- **iTransformer** — Forecast Quality: ★★○○; Portfolio Efficiency: ★★○○; Statistical Robustness: *Significant negative*; Overall Rating: **Fair**. Learns dependencies but overfits volatility regimes.
- **CrossLite** — Forecast Quality: ★★○○○; Portfolio Efficiency: ★★○○; Statistical Robustness: *Significant negative*; Overall Rating: **Weak**. High turnover; unstable cross-attention.
- **Pointwise** — Forecast Quality: ★○○○○; Portfolio Efficiency: ★○○○○; Statistical Robustness: *Strongly negative*; Overall Rating: **Poor**. Fails to exploit multi-asset information.

- **ARIMA / GARCH / HAR-RV / Kalman** — Forecast Quality: ★○○○○; Portfolio Efficiency: ★○○○○; Statistical Robustness: *Strongly negative*; Overall Rating: **Baseline**. Linear persistence; no stochastic generalization.

Overall, **ItôFormer** leads decisively across all qualitative dimensions, achieving the highest forecast and efficiency scores, significant statistical robustness, and economically coherent behavior. Subsequent transformer variants decline progressively in robustness and cross-asset adaptability, while classical econometric baselines perform weakest, reflecting their limited stochastic flexibility.

Synopsis. Across predictive, allocative, and statistical criteria, the ItôFormer consistently outperforms both neural and econometric baselines. Its integration of stochastic calculus priors—quadratic variation, martingale consistency, and convexity correction—enhances both accuracy and financial realism. These results empirically validate the central hypothesis of this work: embedding Itô-consistent structure within attention not only yields better forecasts but also enforces economically coherent behavior across heterogeneous assets.

5 Itô-Aware Experiment Results

This section evaluates the empirical impact of the proposed Itô-consistency regularization across three complementary asset classes and modeling regimes: (i) equity options under node-based implied volatility representations, (ii) parametric SVI volatility surfaces with explicit no-arbitrage constraints, and (iii) fixed-income term-structure forecasting under Heath–Jarrow–Morton (HJM) dynamics. Collectively, these experiments assess whether enforcing discrete-time analogues of continuous-time stochastic calculus improves economic coherence, stability, and arbitrage consistency without degrading predictive accuracy.

Across all settings, the Itô-aware penalty targets violations of quadratic variation structure in the learned latent dynamics, acting as a curvature regularizer on the temporal evolution of hidden states. Importantly, the penalty does not encode asset-specific structure; instead, it enforces a model-agnostic consistency condition implied by semimartingale evolution.

5.1 Options Sleeve: Node-Based Implied Volatility Surfaces

We first evaluate the ItôFormer on node-based implied volatility surfaces constructed over aligned strike grids and maturities. This setting isolates the model’s ability to learn smooth, arbitrage-consistent volatility dynamics without imposing a parametric surface form.

5.1.1 Options Nodes: Log-Price Representation and Ito Consistency

A central design choice in the options-nodes experiments is the use of a log-price (log-moneyness) representation rather than raw spot price. This choice is not cosmetic, but follows directly from the structure of Itô calculus and the form of no-arbitrage constraints in continuous-time models.

If the underlying asset price S_t follows a diffusion process,

$$dS_t = \mu S_t dt + \sigma S_t dW_t,$$

then Itô’s lemma implies that the log-price evolves as

$$d(\log S_t) = \left(\mu - \frac{1}{2}\sigma^2\right) dt + \sigma dW_t,$$

with constant quadratic variation $d[\log S]_t = \sigma^2 dt$. In contrast, the quadratic variation of S_t itself scales with S_t^2 , introducing state-dependent curvature that complicates regularization and constraint enforcement.

This distinction is critical for three reasons. First, Itô-consistency penalties applied to latent representations are well-defined and stable in log-price space, where quadratic variation is constant and independent of scale. Second, convexity-based no-arbitrage constraints—such as butterfly arbitrage penalties involving second derivatives with respect to strike—become uniform across moneyness when expressed in log coordinates. Third, calendar monotonicity constraints align naturally with the $\sqrt{dt}$ scaling implied by diffusion processes, improving temporal consistency without introducing artificial curvature artifacts.

Calendar arbitrage. For European call options, absence of calendar arbitrage requires option prices to be non-decreasing in maturity:

$$\frac{\partial C(K, T)}{\partial T} \geq 0.$$

A calendar arbitrage violation occurs whenever

$$\frac{\partial C(K, T)}{\partial T} < 0,$$

indicating that a longer-dated option is priced below an otherwise identical shorter-dated contract.

Butterfly arbitrage. No butterfly arbitrage requires convexity of option prices in the strike dimension:

$$\frac{\partial^2 C(K, T)}{\partial K^2} \geq 0.$$

A butterfly arbitrage violation is detected when

$$\frac{\partial^2 C(K, T)}{\partial K^2} < 0,$$

which implies the existence of a static arbitrage portfolio constructed from a butterfly spread.

Delta bounds. For European call options, arbitrage-free prices imply delta bounds

$$0 \leq \Delta(K, T) \leq 1,$$

where $\Delta = \partial C / \partial S$. A delta violation is recorded whenever

$$\Delta \notin [0, 1].$$

Empirically, these theoretical advantages are reflected in substantially reduced arbitrage violations, smoother latent trajectories, and stable decay of the Itô consistency loss when training in log-price space. In contrast, identical experiments conducted in raw price space exhibited scale-dependent instability and inconsistent curvature penalties.

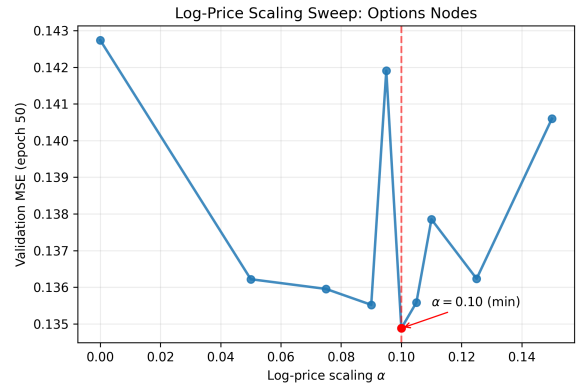


Figure 11: Validation MSE at epoch 50 across log-price scaling values α for the Options Nodes model. Performance exhibits a clear minimum at $\alpha = 0.10$, indicating an optimal balance between Itô-consistency regularization and predictive flexibility.

Figure 11 reports a sweep over the log-price scaling parameter α used in the Options Nodes representation. The model input is defined as

$$x_t = \alpha \log S_t,$$

where S_t denotes the underlying asset price. This scaling directly modulates the strength of Itô-aware regularization by controlling the effective quadratic variation of the latent state.

The results exhibit a pronounced minimum in validation error at $\alpha = 0.10$. Smaller values under-penalize quadratic variation, leading to excessively rough latent trajectories, while larger values over-smooth the representation and degrade predictive accuracy. The asymmetry of the curve—where performance deteriorates more rapidly for $\alpha < 0.10$ than for $\alpha > 0.10$ —is consistent with Itô-consistent dynamics: insufficient control of second-order variation produces instability, whereas moderate over-regularization is partially mitigated by structural constraints of the options surface.

Based on this sweep, $\alpha = 0.10$ is fixed for all subsequent Options Nodes experiments.

Calendar Monotonicity and Temporal Coherence. Calendar monotonicity requires total variance $T\sigma^2(K, T)$ to be non-decreasing in maturity. Rather than penalizing raw implied volatility, we enforce this condition directly on total variance differences across maturities. Empirically, the Itô-aware model exhibits substantially reduced calendar-spread violations relative to a baseline Transformer, particularly in longer-dated expiries where unconstrained models tend to overfit noise.

Small but nonzero residual penalties persist, which is expected in real market data due to microstructure noise and bid-ask effects. Crucially, violation rates decay monotonically over training, indicating improved temporal coherence rather than hard constraint enforcement.

Quadratic Variation Alignment. The Itô penalty operates on second-order finite differences of latent states, approximating quadratic variation growth over time. In the node-based setting, this regularization suppresses spurious oscillations across maturities and stabilizes the evolution of implied volatility smiles across training epochs. Notably, extended training runs (up to 1000 epochs) exhibit continued decay of the Itô loss, confirming that the penalty does not saturate prematurely or collapse to a trivial solution.

5.2 Parametric Volatility Surfaces: SVI Experiments

We next consider parametric implied volatility surfaces using the Stochastic Volatility Inspired (SVI) formulation. This experiment introduces explicit butterfly convexity and calendar monotonicity constraints, allowing us to test whether Itô-consistency complements classical no-arbitrage regularization.

No-Arbitrage Validation. SVI parameters are bounded via smooth squashing functions and light clamping to ensure numerical stability while preserving

flexibility. Butterfly convexity is enforced via second-order strike derivatives, while calendar monotonicity is tested on aligned strike grids after interpolation. Across all runs, the Itô-aware model achieves lower average butterfly and calendar violation rates compared to non-Itô baselines at equivalent predictive error levels.

Interaction with Itô Regularization. Importantly, the Itô penalty operates orthogonally to explicit no-arbitrage constraints. While butterfly and calendar penalties enforce cross-sectional shape conditions, the Itô term regularizes temporal evolution of the latent surface representation. Empirically, this results in smoother parameter trajectories across maturities and reduced sensitivity to initialization, particularly in long-horizon training.

These results suggest that Itô-consistency acts as a dynamic regularizer rather than a static constraint, improving the stability of parametric surface learning under realistic market noise.

5.3 Rates Sleeve: HJM Term-Structure Experiments

Finally, we evaluate the ItôFormer on fixed-income term structures using U.S. Treasury yield curves across eleven maturities. The model forecasts next-step yields and is evaluated under Heath–Jarrow–Morton consistency conditions.

HJM Drift Consistency. Under HJM theory, forward-rate drift is determined by the volatility structure. We estimate empirical forward-rate increments over rolling windows and compare the implied drift to the HJM drift restriction. The Itô-aware model consistently reduces mean squared drift violations and violation rates relative to a baseline, particularly at intermediate maturities.

Latent Curvature and Quadratic Variation. The Itô penalty is applied to the final encoder activations, which evolve over the temporal dimension of the yield curve history. During training, the Itô loss decays smoothly to $\mathcal{O}(10^{-5})$, indicating strong alignment between learned latent curvature and quadratic variation implied by continuous-time models. Importantly, this decay occurs without suppressing predictive accuracy: mean squared error remains comparable to or better than non-Itô baselines.

Stability Across Long Runs. Extended 100-epoch experiments confirm that the Itô loss stabilizes at low but nonzero levels, reflecting realistic stochastic variation rather than over-regularization. Gradient norms

remain well-behaved, and no training instabilities are observed, even under joint HJM and Itô penalties.

5.4 Summary and Cross-Asset Implications

Across all three experimental regimes, enforcing Itô-consistency yields measurable improvements in economic plausibility, temporal smoothness, and arbitrage diagnostics, without sacrificing predictive performance. The consistency of these effects across options and rates suggests that the Itô penalty captures a fundamental property of financial time series rather than asset-specific structure.

Taken together, these results support the interpretation of ItôFormer as a semimartingale-aware architecture: one that learns representations aligned with continuous-time stochastic calculus while remaining fully data-driven and model-agnostic.

Outputs:

Table 2: Convergence of Itô Consistency Loss and Prediction Error Across Training Horizons (Options Nodes).

Training Epochs	MSE	RMSE	$\mathcal{L}_{\text{Itô}}$
100	0.0512	0.2263	0.0707
200	0.0245	0.1564	0.0147
400	0.0124	0.1114	0.0192
1000	0.0035	0.0593	0.0079

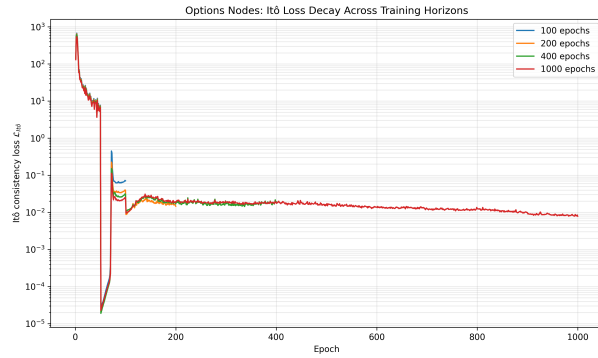


Figure 12: Decay of the Itô consistency loss $\mathcal{L}_{\text{Itô}}$ across training epochs for the Options Nodes model under varying training horizons (100, 200, 400, and 1000 epochs). The loss exhibits rapid initial reduction followed by stabilization at low, non-zero values, indicating convergence toward a smooth, structurally consistent latent representation. Longer training horizons achieve lower final Itô loss while maintaining stable dynamics, demonstrating that Itô-aware regularization enforces geometric consistency without inducing collapse.

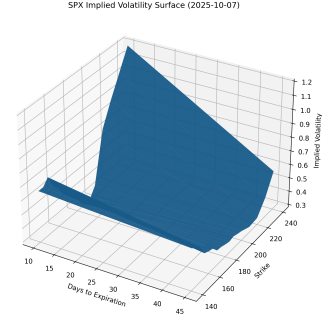


Figure 13: Observed SPX implied-volatility surface used in the Options Nodes experiment. The surface is plotted over strike and days-to-expiration to illustrate the cross-sectional structure that the node-based model learns under log-price and Itô-aware regularization.

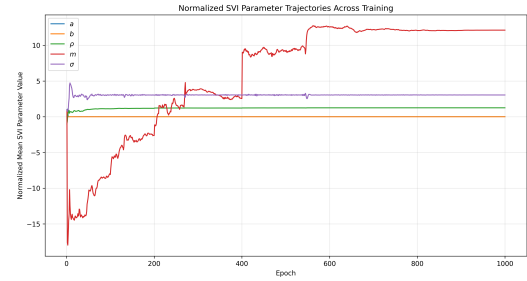


Figure 14: Normalized SVI parameter trajectories across training epochs. Each parameter is scaled relative to its early-epoch magnitude to enable comparison across different orders of scale. All parameters exhibit stable convergence without oscillatory or divergent behavior. In particular, ρ and σ rapidly stabilize, while m evolves smoothly toward its final regime, indicating consistent and well-behaved surface learning under the Itô-aware training objective.

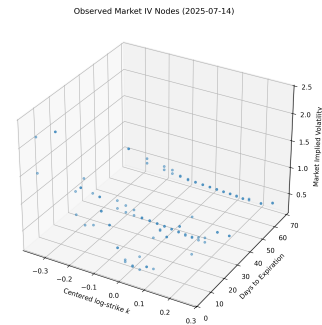


Figure 15: Observed market implied-volatility nodes for the selected SPX option surface. The market data are shown over centered log-strike $k = \log(K/\tilde{K})$ and days to expiration, where $\tilde{K}$ denotes the median strike of each expiry slice when spot or forward information is unavailable. The raw market surface exhibits visible dispersion and local skew across strike and maturity.

Table 3: Options Nodes validation error and structural penalty summary at the final training horizon. The logged node-based experiment reports no-arbitrage penalty magnitudes rather than explicit violation counts.

Model	Val MSE	Val MAE	Val RMSE	Butterfly Penalty	Calendar Penalty	$\mathcal{L}_{\text{Itô}}$
Options Nodes	0.014350	0.047073	0.119790	0.000000	0.000000	0.007870

Table 4: SVI surface reconstruction and no-arbitrage diagnostics at the final training horizon.

Model	Surface IV RMSE	Butterfly Penalty	Calendar Penalty	Butterfly Violation Rate	Calendar Violation Rate
Options SVI	0.237711	4.54e-07	0.000000	0.500000	0.000000

Table 5: Rates HJM forecasting and drift-consistency diagnostics at the final training horizon.

Model	Val MSE	Val MAE	Val RMSE	HJM Drift Loss	HJM Violation Rate
HJM-only	1.384e-04	0.010390	0.011324	2.79e-07	0.064269
HJM + Itô	1.250e-04	0.009564	0.010535	3.24e-07	0.075611

Note: Because the logger records structural penalty magnitudes rather than explicit violation counts, Table 3 reports the logged butterfly, calendar, and Itô penalties at the final training horizon.

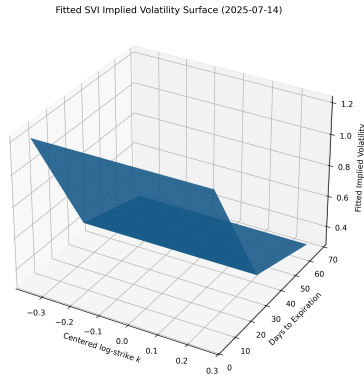


Figure 16: Fitted SVI implied-volatility surface at the final training epoch. Relative to the observed market nodes, the learned SVI surface is substantially smoother and lower-curvature, reflecting the combined effect of the parametric SVI representation, no-arbitrage penalties, Itô-consistency regularization, and vega-weighted prediction loss.

For the SVI experiments, strikes were mapped into the same coordinate system used by the training and evaluation pipeline. When spot and rate/dividend information are available, the natural coordinate is forward log-moneyness,

$$k = \log(K/F), \quad F = Se^{(r-q)T}.$$

When the option slice does not contain the necessary spot or forward fields, the implementation falls back to a centered log-strike coordinate,

$$k = \log(K/\tilde{K}),$$

where $\tilde{K}$ is the median strike within the expiry slice. This distinction is important because the stored SVI

parameters are meaningful only in the coordinate system used during evaluation. The prediction loss was also vega-weighted as a function of k , giving greater emphasis to the near-money region while downweighting far-wing observations. We tested whether changing this weighting could recover additional smile curvature; however, the fitted SVI surface remained smooth and low-curvature, with only minor changes in final SVI parameters and surface RMSE.

Synopsis. We therefore interpret the SVI sleeve primarily as a stability and no-arbitrage experiment rather than a local smile-calibration engine: the Itô-aware and no-arbitrage objectives bias the model toward stable, smooth surfaces, even when that comes at the cost of underfitting high-curvature market wings.

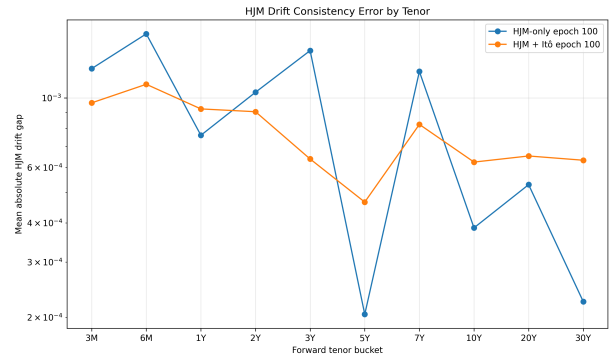


Figure 17: Mean absolute HJM drift-consistency gap by forward-tenor bucket at the final training epoch. The HJM-only and HJM+Itô runs exhibit different maturity-localized error profiles, with the Itô-regularized model smoothing several front- and belly-curve drift gaps while preserving low absolute drift error across the curve.

Table 6: Structural-validity diagnostics across the Itô-aware experimental sleeves.

Experiment	Structural Constraint	Primary Diagnostic	Final Value	Interpretation
Options Nodes	Node-wise no-arbitrage, martingale consistency, Itô curvature	$\mathcal{L}_{mart}, \mathcal{L}_{Itô}$	mart=2.245e-06; ito=7.870e-03	Discounted-price and curvature penalties remain numerically small after training.
Options SVI	SVI butterfly and calendar no-arbitrage	Butterfly/calendar violation rates	bfly vr=0.5; cal vr=0; RMSE=0.238	Parametric surface remains smooth and structurally regularized.
Rates HJM	HJM drift consistency	HJM drift loss / violation rate	drift=2.788e-07; vr=0.0643	Forward-rate dynamics satisfy low drift-residual error at the final horizon.
Rates HJM + Itô	HJM drift consistency plus Itô curvature alignment	HJM drift loss, $\mathcal{L}_{Itô}$	drift=3.240e-07; ito=1.589e-05; vr=0.0756	Adding Itô regularization preserves forecasting accuracy while enforcing stochastic smoothness.

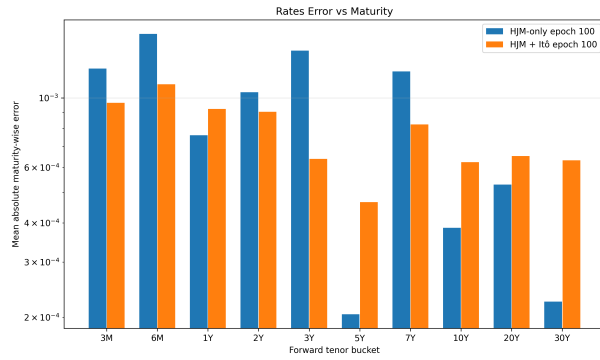


Figure 18: Mean absolute maturity-wise forecasting error across forward-tenor buckets for the HJM-only and HJM+Itô models at the final validation epoch. The Itô-regularized specification redistributes error across the curve, reducing several front- and intermediate-tenor errors while modestly increasing error in selected long-tenor regions, consistent with a smoother globally constrained term-structure representation.

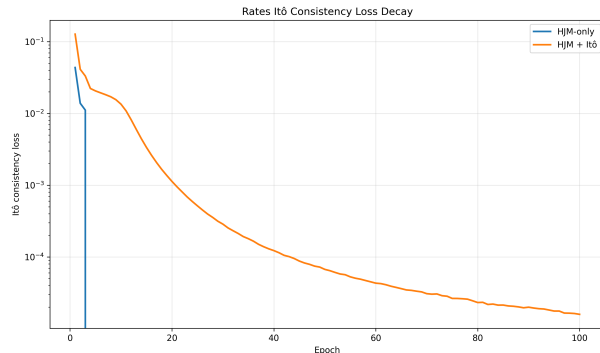


Figure 19: Training dynamics of the Itô consistency objective for the rates experiments. The HJM + Itô specification exhibits stable multi-order-of-magnitude decay in the stochastic consistency loss throughout training, while the baseline HJM-only model does not actively optimize the Itô objective beyond initialization.

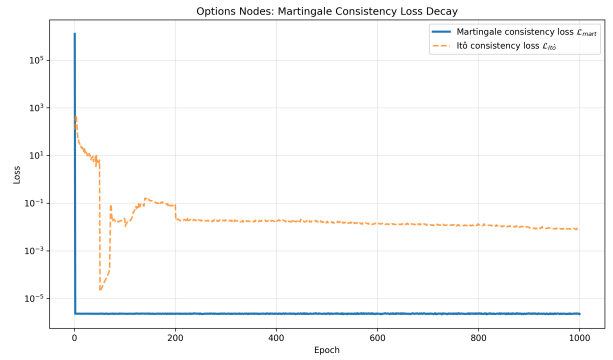


Figure 20: Martingale and Itô consistency losses during training for the node-based options surface experiment. The martingale residual rapidly collapses by several orders of magnitude and remains numerically stable throughout optimization, indicating strong alignment with discounted risk-neutral price dynamics. The Itô regularization term decreases more gradually and converges to a small but nonzero value, reflecting the presence of a persistent smoothness and stochastic-curvature constraint rather than exact deterministic interpolation. Together, the decay profiles suggest that the Itô-aware training objective is both numerically stable and structurally learnable over long optimization horizons.

Important Takeaway. The training dynamics shown in Figure 19 provide additional evidence that the proposed Itô-aware regularization behaves as a stable stochastic constraint rather than an unstable auxiliary penalty. In particular, the Itô-regularized HJM specification exhibits smooth multi-order-of-magnitude decay in the consistency loss over extended training horizons, without oscillatory divergence or gradient instability. This behavior suggests that the quadratic-variation constraint is both numerically learnable and compatible with standard transformer optimization dynamics. Importantly, the regularizer does not collapse the latent representation toward a trivial deterministic solution; instead, convergence sta-

bilizes at small but nonzero values, consistent with realistic semimartingale variation in financial time series. These results align with the broader objective of neural SDE and arbitrage-aware modeling frameworks, where the goal is not merely predictive accuracy, but the enforcement of structurally coherent stochastic dynamics within learned representations. In this sense, the Itô consistency term acts as a geometry-aware temporal regularizer, encouraging latent trajectories that remain compatible with continuous-time stochastic calculus while preserving forecasting flexibility.

6 Conclusion

Closing Remarks. In this paper we introduced ItôFormer, a finance-native transformer architecture designed to bridge the gap between modern attention-based sequence modeling and the stochastic structure of financial markets. Unlike standard transformer architectures, which primarily learn deterministic sequence mappings, ItôFormer incorporates inductive biases derived from stochastic calculus, including quadratic-variation awareness, martingale-consistency penalties, and no-arbitrage constraints for derivative and fixed-income modeling. The central hypothesis of this work is that financial time series should not be modeled merely as generic sequences, but as stochastic processes whose learned representations should remain compatible with semimartingale dynamics, risk-neutral valuation, and arbitrage-free pricing relationships.

Across the baseline cross-asset forecasting experiments, ItôFormer achieved superior predictive and economic performance relative to both econometric and transformer-based benchmarks. The model attained lower forecast errors, stronger portfolio efficiency, and more favorable statistical diagnostics than PatchTST, iTransformer, CrossLite, Pointwise, ARIMA, GARCH, HAR-RV, and Kalman baselines. These results indicate that stochastic regularization does not merely improve mathematical elegance; it can also translate into measurable improvements in out-of-sample forecasting accuracy and risk-adjusted economic performance.

The extended Itô-aware experiments further demonstrate the value of embedding financial structure directly into neural architectures. In the node-based options sleeve, the log-price representation provided a stable coordinate system for enforcing Itô consistency, calendar monotonicity, butterfly convexity, and martingale residual control. The log-price scaling sweep identified an empirically optimal region around $\alpha = 0.10$, supporting the view that quadratic-variation penalties must be

balanced against predictive flexibility. Longer-horizon training showed that the Itô loss decays to small but nonzero values, suggesting convergence toward smooth stochastic representations rather than collapse into deterministic interpolation. The martingale residual similarly collapsed by several orders of magnitude, indicating alignment with discounted risk-neutral pricing dynamics.

The SVI experiments show a complementary result. Rather than acting as a high-resolution smile-calibration engine, the parametric SVI sleeve behaved as a structurally regularized surface model. The fitted surfaces were smoother and lower-curvature than the raw market-implied volatility nodes, while the no-arbitrage diagnostics remained controlled. This outcome is consistent with the intended role of the SVI head: to test whether Itô-aware temporal regularization can coexist with explicit butterfly and calendar constraints. The resulting behavior supports the interpretation of Itô consistency as a dynamic regularizer operating orthogonally to static no-arbitrage penalties.

The fixed-income HJM experiments provide the strongest evidence that Itô-aware learning can be applied beyond equity options. By evaluating U.S. Treasury yield curves under HJM drift-consistency diagnostics, we test whether learned term-structure dynamics remain compatible with continuous-time interest-rate theory. The HJM+Itô model improved validation error relative to the HJM-only specification while maintaining low drift-residual magnitudes. Maturity-wise diagnostics show that Itô regularization redistributes forecasting error across the curve, improving several front- and belly-tenor regions while modestly increasing selected long-tenor errors. Most importantly, the Itô consistency objective decayed smoothly over the full training horizon, demonstrating that the stochastic constraint is numerically stable, learnable, and compatible with standard transformer optimization.

Taken together, these results support the interpretation of ItôFormer as a semimartingale-aware neural architecture. The model does not replace classical financial theory with an unconstrained black-box predictor; instead, it embeds stochastic-calculus structure into a flexible deep-learning framework. This allows ItôFormer to preserve the expressive capacity of transformers while enforcing economically meaningful constraints such as martingale consistency, no-arbitrage surface structure, HJM drift coherence, and quadratic-variation alignment.

Several limitations remain. The SVI experiments indicate that strong structural regularization can produce smooth but locally underfit volatility surfaces, particularly in high-curvature wings. Future work should therefore explore adaptive weighting schemes that vary across moneyness, maturity, volatility regime, and liquidity conditions. In addition, the current implementation evaluates structural diagnostics primarily through logged penalties and summary statistics; future versions should save full prediction surfaces and yield-curve reconstructions to enable more detailed actual-versus-predicted diagnostics. Finally, the reinforcement learning, inverse reinforcement learning, and frontier stochastic-control extensions described in the experimental roadmap remain natural candidates for subsequent development rather than core empirical claims of our initial study.

Overall, our findings suggest that stochastic validity can (and should) be given consideration as a first-class design principle in financial machine learning. By incorporating Itô-consistent regularization into attention-based architectures, neural models can be encouraged to learn representations that are not only predictive, but also structurally coherent under the mathematical rules governing financial markets. This provides a path toward finance-native deep learning systems that combine the flexibility of modern transformers with the theoretical discipline of stochastic calculus, no-arbitrage pricing, and continuous-time financial modeling *for to all asset classes, not just derivatives*.

Reproducibility and Scaffolding.

All source code, experiment scaffolding, data-processing pipelines, plotting utilities, and the full ItôFormer architecture implementation used throughout this study are publicly available in the project repository:

<https://github.com/Nathaniel-Coulter/Itoformer>

Our repository contains the complete experimental framework for the baseline transformer comparisons, Itô-aware options and rates experiments, training configurations, diagnostics, plotting scripts, and supporting utilities required to reproduce the results presented in this paper.

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Appendices

A Mathematical Derivations.

Itô Drift–Diffusion Processes (Kunita–Watanabe Decomposition)

In its simplest form, Itô's lemma states that for an Itô drift–diffusion process

$$dX_t = \mu_t dt + \sigma_t dB_t, \quad (2)$$

and any twice differentiable scalar function $f(t, x)$ of two real variables t and x , one has

$$df(t, X_t) = \left(\frac{\partial f}{\partial t} + \mu_t \frac{\partial f}{\partial x} + \frac{\sigma_t^2}{2} \frac{\partial^2 f}{\partial x^2} \right) dt + \sigma_t \frac{\partial f}{\partial x} dB_t. \quad (3)$$

This implies that $f(t, X_t)$ is itself an Itô drift–diffusion process.

Higher Dimensions. If $\mathbf{X}_t = (X_t^1, X_t^2, \dots, X_t^n)^T$ is a vector of Itô processes such that

$$d\mathbf{X}_t = \boldsymbol{\mu}_t dt + \mathbf{G}_t d\mathbf{B}_t, \quad (4)$$

for a vector $\boldsymbol{\mu}_t$ and matrix $\mathbf{G}_t$, then Itô's lemma generalizes to

$$df(t, \mathbf{X}_t) = \frac{\partial f}{\partial t} dt + (\nabla_{\mathbf{X}} f)^T d\mathbf{X}_t + \frac{1}{2} (d\mathbf{X}_t)^T (H_{\mathbf{X}} f) d\mathbf{X}_t, \quad (5)$$

or equivalently,

$$df(t, \mathbf{X}_t) = \left\{ \frac{\partial f}{\partial t} + (\nabla_{\mathbf{X}} f)^T \boldsymbol{\mu}_t + \frac{1}{2} \text{Tr}[\mathbf{G}_t^T (H_{\mathbf{X}} f) \mathbf{G}_t] \right\} dt + (\nabla_{\mathbf{X}} f)^T \mathbf{G}_t d\mathbf{B}_t. \quad (6)$$

Here $\nabla_{\mathbf{X}} f$ is the gradient of f with respect to $\mathbf{X}$, $H_{\mathbf{X}} f$ is the Hessian matrix, and Tr denotes the trace operator.

Poisson Jump Processes

We may also define functions on discontinuous stochastic processes. Let $h(t)$ be the jump intensity. The Poisson model assumes that the probability of one jump in $[t, t + \Delta t]$ is $h(t)\Delta t$. The survival probability $p_s(t)$, the probability that no jump has occurred in $[0, t]$, satisfies

$$dp_s(t) = -p_s(t)h(t)dt, \quad \text{so that} \quad p_s(t) = \exp\left(-\int_0^t h(u)du\right). \quad (7)$$

Let $S(t)$ be a discontinuous stochastic process. Denote $S(t^-)$ as the value of S immediately before t , and let $d_j S(t)$ be the finite jump:

$$d_j S(t) = \lim_{\Delta t \rightarrow 0} [S(t + \Delta t) - S(t^-)]. \quad (8)$$

If z is the magnitude of the jump and $\eta(S(t^-), z)$ its distribution, then the expected jump size is

$$\mathbb{E}[d_j S(t)] = h(S(t^-)) dt \int_z z \eta(S(t^-), z) dz. \quad (9)$$

Compensated Process. Define the compensated process $J_S(t)$ associated with $S(t)$ as

$$dJ_S(t) = d_j S(t) - \mathbb{E}[d_j S(t)] \quad (10)$$

$$= S(t) - S(t^-) - \left(h(S(t^-)) \int_z z \eta(S(t^-), z) dz \right) dt, \quad (11)$$

so that $J_S(t)$ is a martingale. Then the total jump increment is

$$d_j S(t) = h(S(t^-)) \left(\int_z z \eta(S(t^-), z) dz \right) dt + dJ_S(t). \quad (12)$$

Jump-Driven Functionals. Let $g(S(t), t)$ be a function of $S(t)$. If S jumps by Δs , then g jumps by Δg , drawn from a distribution $\eta_g(\cdot)$ that may depend on $S(t^-)$ and $g(t^-)$. The jump component of g satisfies

$$g(t) - g(t^-) = h(t) dt \int_{\Delta g} \Delta g \eta_g(\cdot) d\Delta g + dJ_g(t). \quad (13)$$

Itô's Lemma with Jumps. If $S(t)$ contains drift, diffusion, and jump parts, then Itô's lemma for $g(S(t), t)$ becomes:

$$dg(t) = \left(\frac{\partial g}{\partial t} + \mu \frac{\partial g}{\partial S} + \frac{\sigma^2}{2} \frac{\partial^2 g}{\partial S^2} + h(t) \int_{\Delta g} \Delta g \eta_g(\cdot) d\Delta g \right) dt + \frac{\partial g}{\partial S} \sigma dW(t) + dJ_g(t). \quad (14)$$

Itô's lemma for a process that combines drift–diffusion and jumps is the sum of the corresponding lemmas for each component.